

PRESENTATION TOPIC:

Bipolar Junction Transistors

Bipolar Junction Transistors

- The transistor is a three-layer semiconductor device consisting of either two n- and one p-type layers of material or two p- and one n-type layers of material.
- The former is called an npn transistor, while the latter is called a pnp transistor
- So, there are two types of BJT-
 - i) pnp transistor
 - ii) npn transistor

Bipolar Junction Transistors

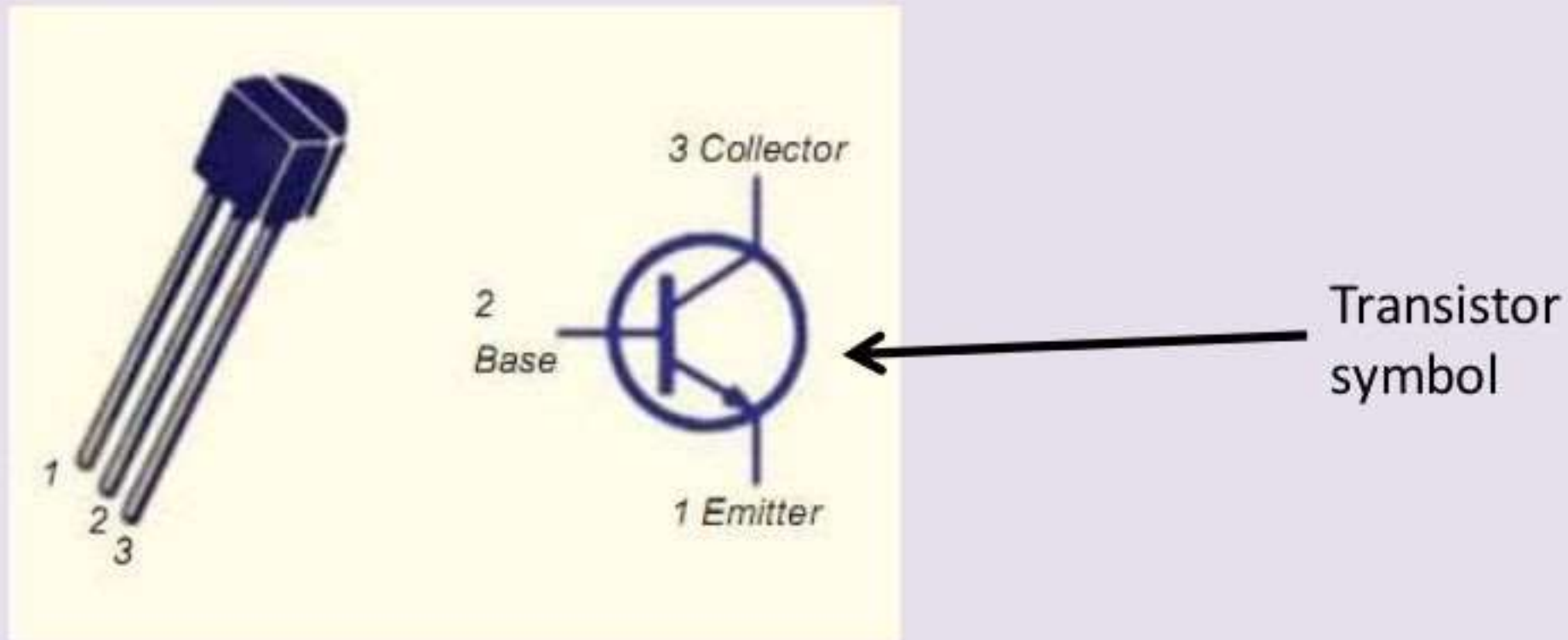


In each transistor following points to be noted-

- i) There are two junction, so transistor can be considered as two diode connected back to back.
- ii) There are three terminals.
- iii) The middle section is thin than other.

Naming of Transistor Terminals

- Transistor has three section of doped semiconductor.
- The section one side is called “emitter” and the opposite side is called “collector”.
- The middle section is called “base”.



Naming of Transistor Terminals

1) Emitter:

→ The section of one side that supplies carriers is called emitter.

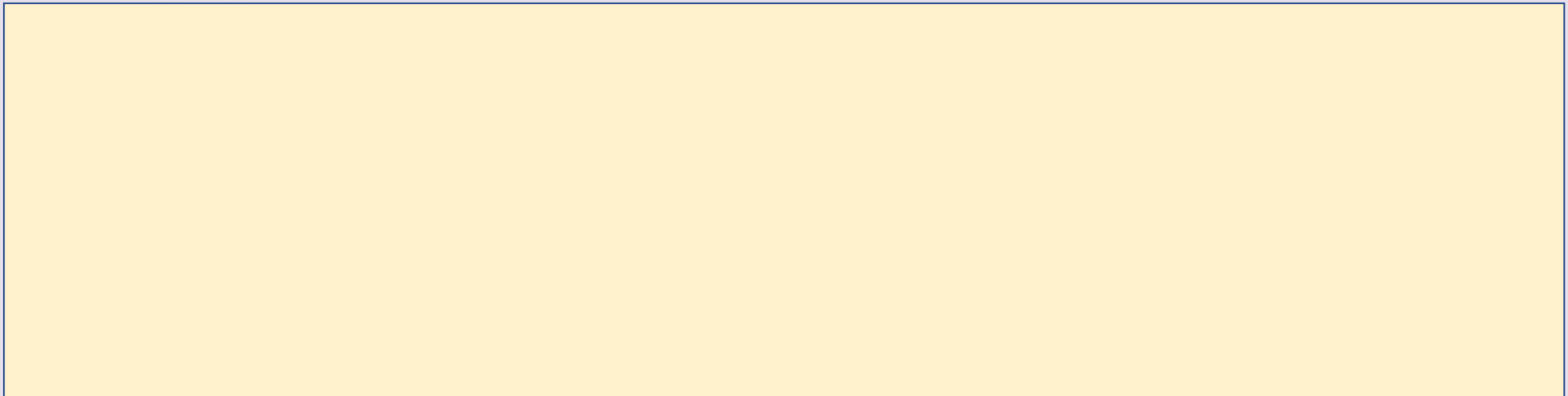
→ Emitter is always forward biased wr to base so it can supply carrier.



Naming of Transistor Terminals

2) Collector:

- The section on the other side that collects carrier is called collector.
- The collector is always reversed biased wr to base.



Naming of Transistor Terminals

3) Base:

→ The middle section which forms two pn junction between emitter and collector is called Base.

Some important factors to be remembered-

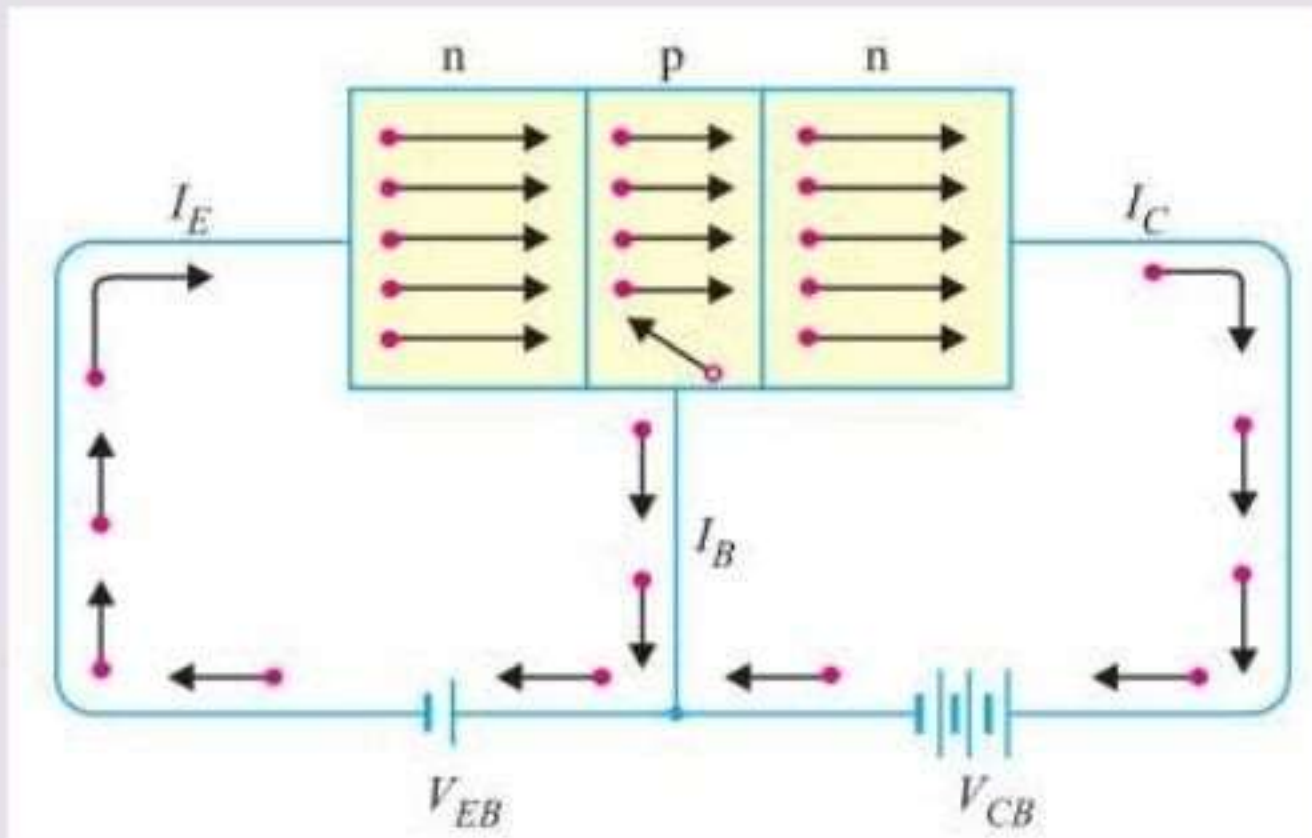
- The transistor has three region named emitter, base and collector.
- The Base is much thinner than other region.
- Emitter is heavily doped so it can inject large amount of carriers into the base.
- Base is lightly doped so it can pass most of the carrier to the collector.
- Collector is moderately doped.

Some important factors to be remembered-

- The junction between emitter and base is called emitter-base junction(emitter diode) and junction between base and collector is called collector-base junction(collector diode).
- The emitter diode is always forward biased and collector diode is reverse biased.
- The resistance of emitter diode is very small(forward) and resistance of collector diode is high(reverse).

Transistor Operation

1) Working of npn transistor:



✓ Forward bias is applied to emitter-base junction and reverse bias is applied to collector-base junction.

✓ The forward bias in the emitter-base junction causes electrons to move toward base. This constitutes emitter current, I_E

Transistor Operation

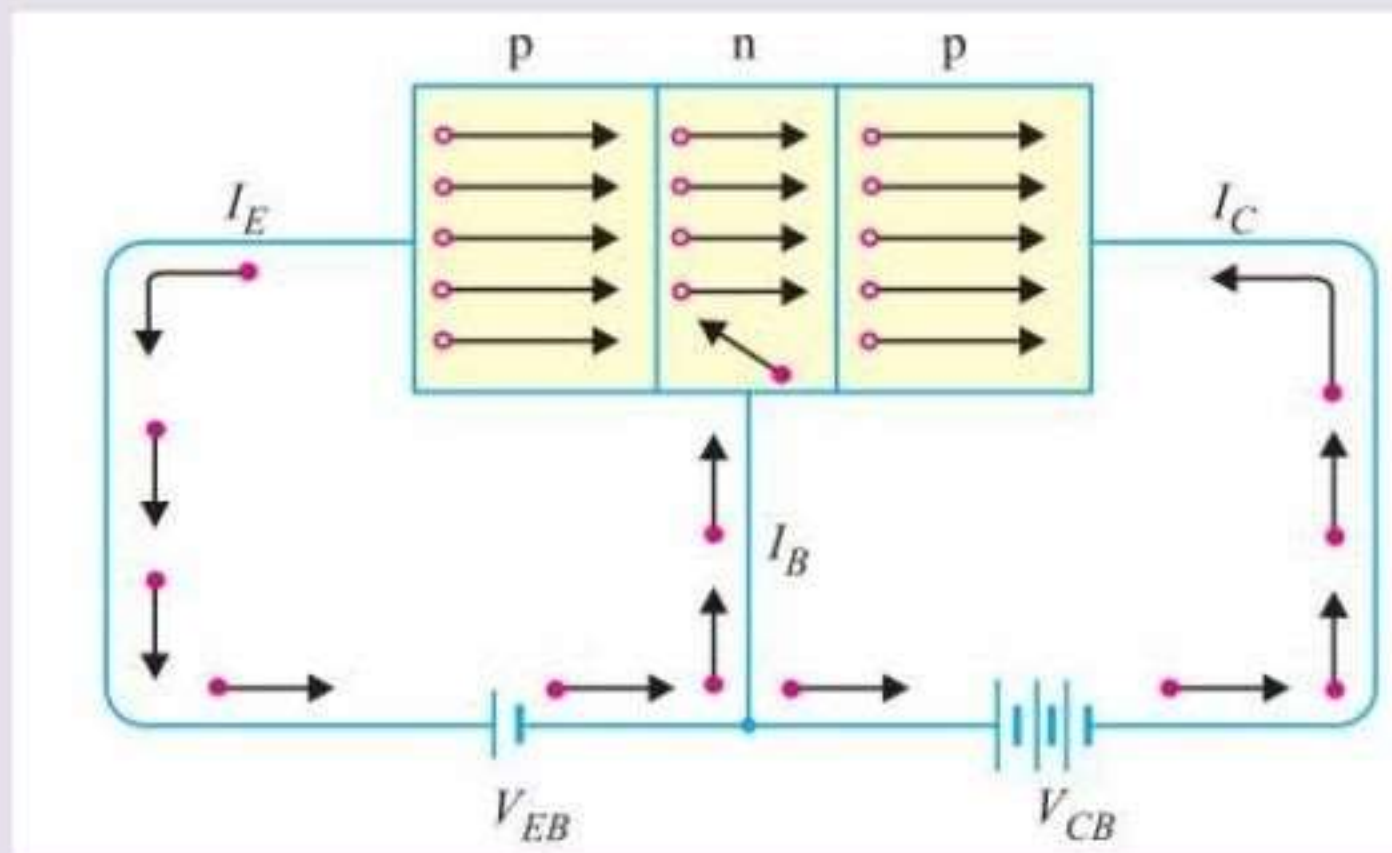
1) Working of npn transistor:

- ✓ As this electrons flow toward p-type base, they try to recombine with holes. As base is lightly doped only few electrons recombine with holes within the base.
- ✓ These recombined electrons constitute small base current.
- ✓ The remainder electrons crosses base and constitute collector current.

$$I_E = I_B + I_C$$

Transistor Operation

2) Working of pnp transistor:



✓ Forward bias is applied to emitter-base junction and reverse bias is applied to collector-base junction.

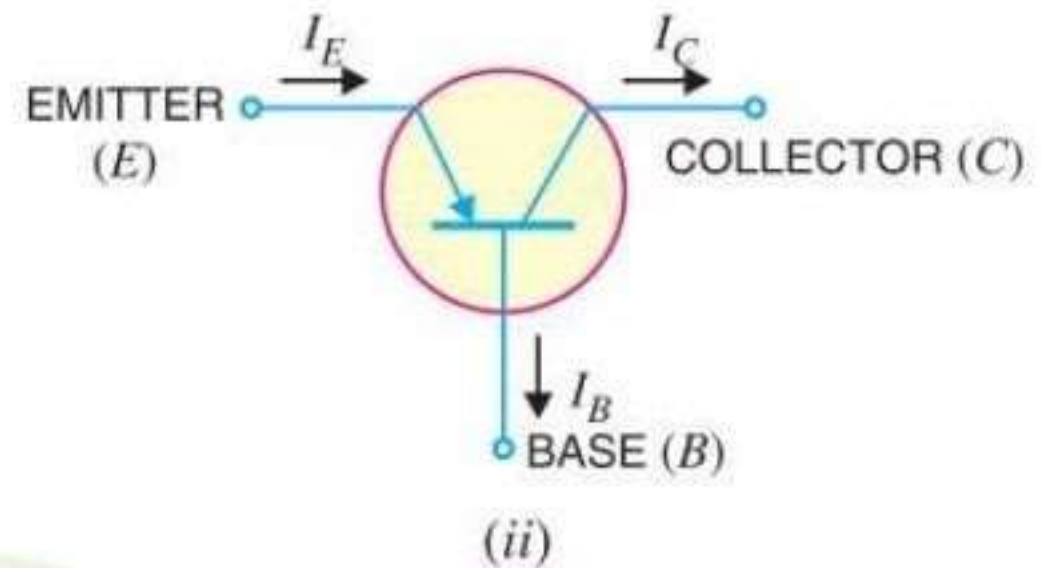
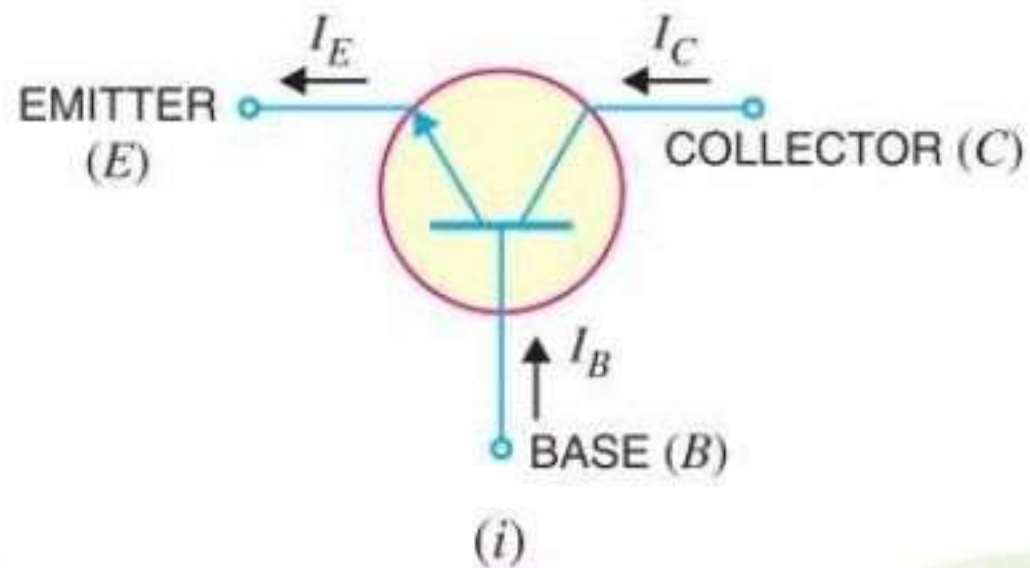
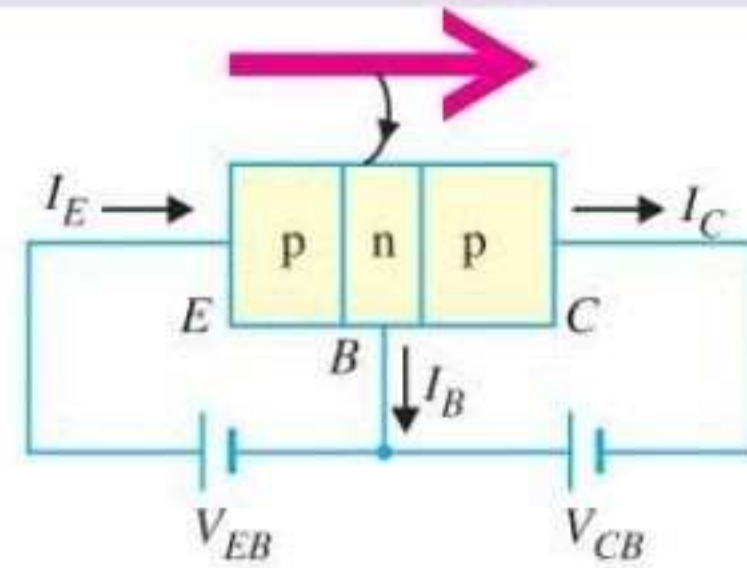
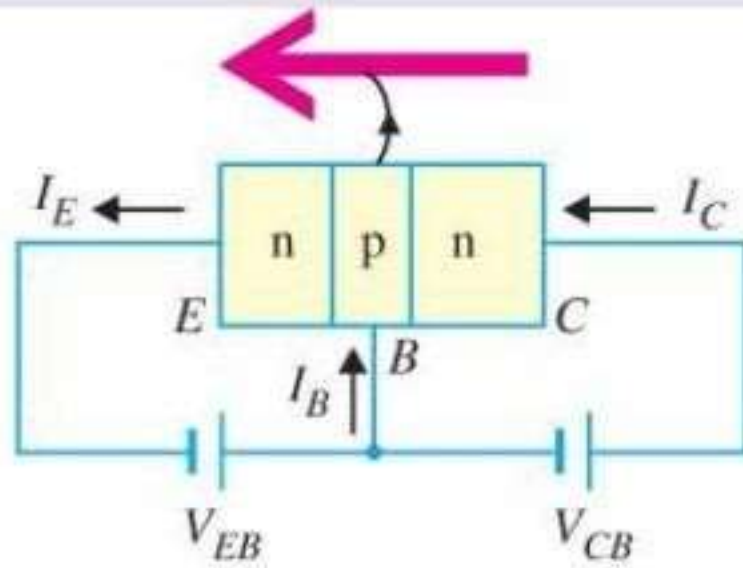
✓ The forward bias in the emitter-base junction causes holes to move toward base. This constitute emitter current, I_E

Transistor Operation

2) Working of pnp transistor:

- ✓ As this holes flow toward n-type base, they try to recombine with electrons. As base is lightly doped only few holes recombine with electrons within the base.
- ✓ These recombined holes constitute small base current.
- ✓ The remainder holes crosses base and constitute collector current.

Transistor Symbol



Transistor Operating Modes

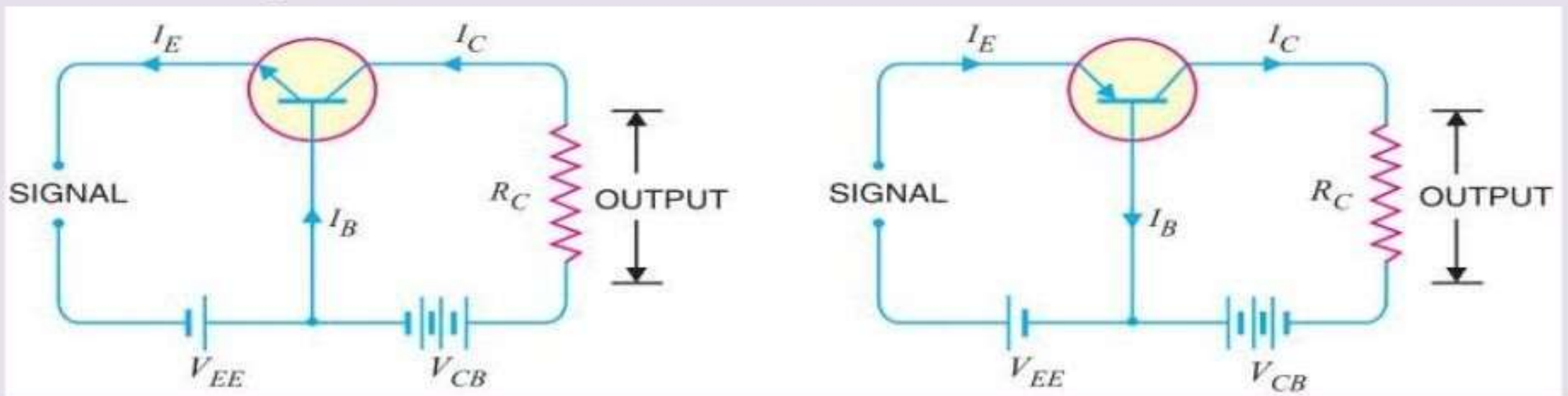
- Active Mode
 - Base- Emitter junction is forward and Base-Collector junction is reverse biased.
- Saturation Mode
 - Base- Emitter junction is forward and Base-Collector junction is forward biased.
- Cut-off Mode
 - Both junctions are reverse biased.

Transistor Connection

- Transistor can be connected in a circuit in following three ways-
 - 1) Common Base
 - 2) Common Emitter
 - 3) Common Collector

Common Base Connection

- The common-base terminology is derived from the fact that the base is common to both the input and output sides of the configuration.



- First Figure shows common base npn configuration and second figure shows common base pnp configuration.

Common Base Connection

- Current amplification factor (α) :

The ratio of change in collector current to the change in emitter current at constant V_{CB} is known as current amplification factor, α .

$$\alpha = \frac{\Delta I_C}{\Delta I_E} \text{ at constant } V_{CB}$$

→ Practical value of α is less than unity, but in the range of 0.9 to 0.99

Expression for Collector Current

→ Total emitter current does not reach the collector terminal, because a small portion of it constitute base current. So,

$$I_E = I_C + I_B$$

→ Also, collector diode is reverse biased, so very few minority carrier passes the collector-base junction which actually constitute leakage current, I_{CBO} .

→ So, collector current constitute of portion of emitter current αI_E and leakage current I_{CBO} .

$$I_C = \alpha I_E + I_{CBO}$$

Expression for Collector Current

$$\begin{aligned}I_C &= \alpha (I_C + I_B) + I_{CBO} \\I_C (1 - \alpha) &= \alpha I_B + I_{CBO} \\I_C &= \frac{\alpha}{1 - \alpha} I_B + \frac{I_{CBO}}{1 - \alpha}\end{aligned}$$

8.9 Characteristics of Common Base Connection

The complete electrical behaviour of a transistor can be described by stating the interrelation of the various currents and voltages. These relationships can be conveniently displayed graphically and the curves thus obtained are known as the characteristics of transistor. The most important characteristics of common base connection are *input characteristics* and *output characteristics*.

1. Input characteristic. It is the curve between emitter current I_E and emitter-base voltage V_{EB} at constant collector-base voltage V_{CB} . The emitter current is generally taken along y -axis and emitter-base voltage along x -axis. Fig. 8.14 shows the input characteristics of a typical transistor in CB arrangement. The following points may be noted from these characteristics :

(i) The emitter current I_E increases rapidly with small increase in emitter-base voltage V_{EB} . It means that input resistance is very small.

(ii) The emitter current is almost independent of collector-base voltage V_{CB} . This leads to the conclusion that emitter current (and hence collector current) is almost independent of collector voltage.

Input resistance. It is the ratio of change in emitter-base voltage (ΔV_{EB}) to the resulting

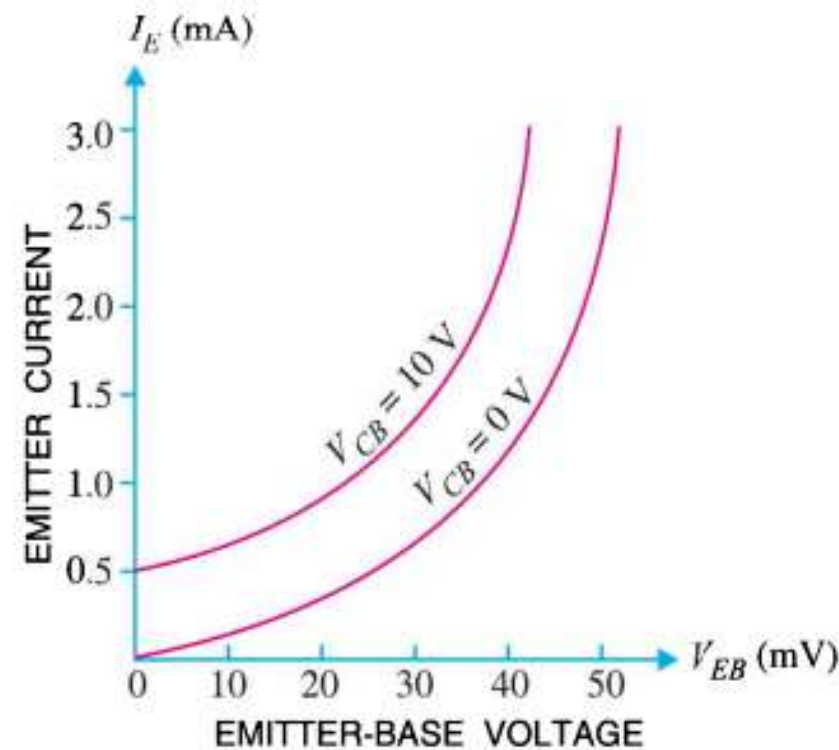


Fig. 8.14

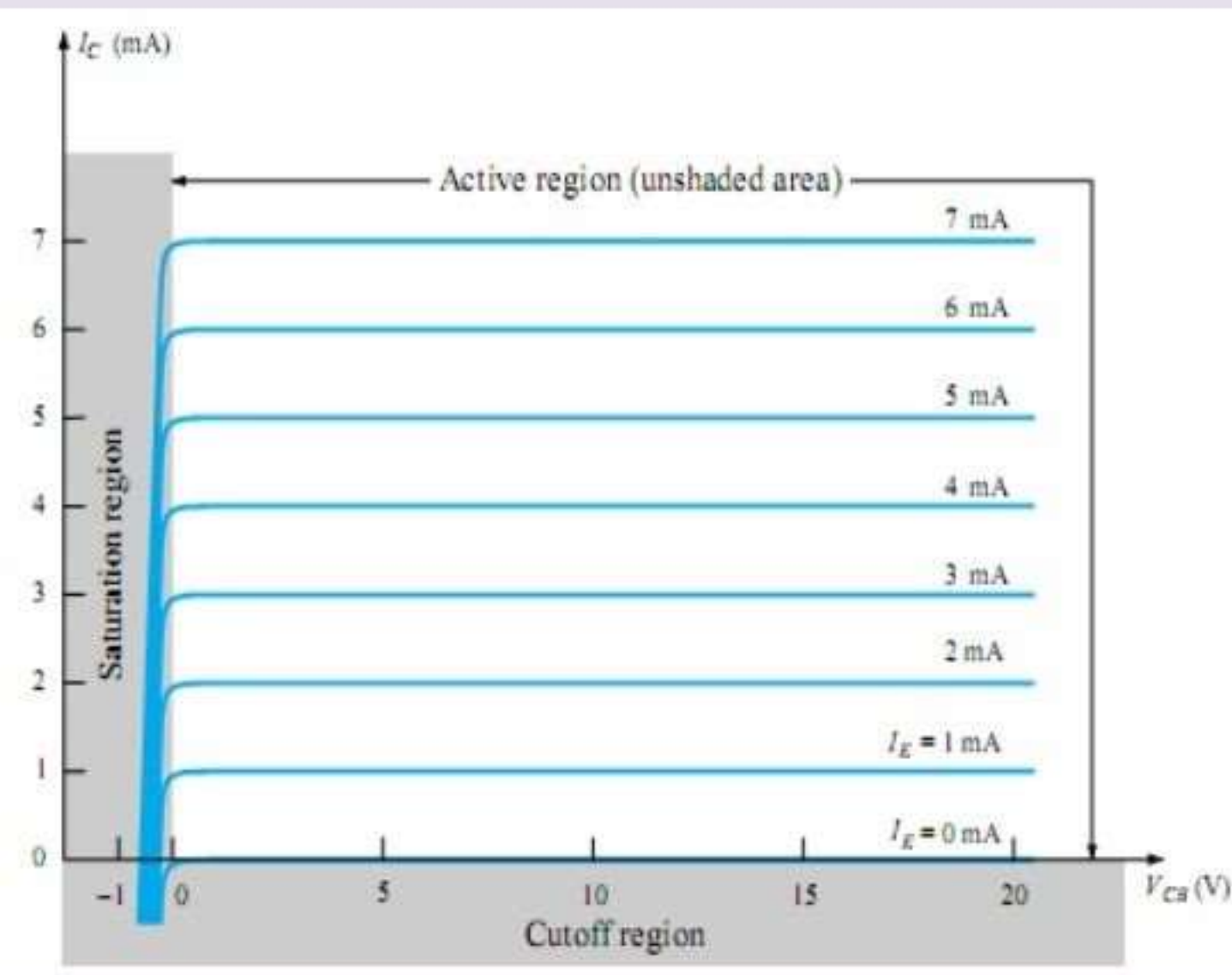
change in emitter current (ΔI_E) at constant collector-base voltage (V_{CB}) i.e.

$$\text{Input resistance, } r_i = \frac{\Delta V_{BE}}{\Delta I_E} \text{ at constant } V_{CB}$$

In fact, input resistance is the opposition offered to the signal current. As a very small V_{EB} is sufficient to produce a large flow of emitter current I_E , therefore, input resistance is quite small, of the order of a few ohms.

Characteristics of common base configuration

Output Characteristics:



→ V_{BC} vs I_C

characteristics is called output characteristics.

→ I_C varies linearly with V_{BC} , only when V_{BC} is very small.

→ As, V_{BC} increases, I_C becomes constant.

Input and Output Resistance of common base conf.

- Input Resistance: The ratio of change in emitter-base voltage to the change in emitter current is called Input Resistance.

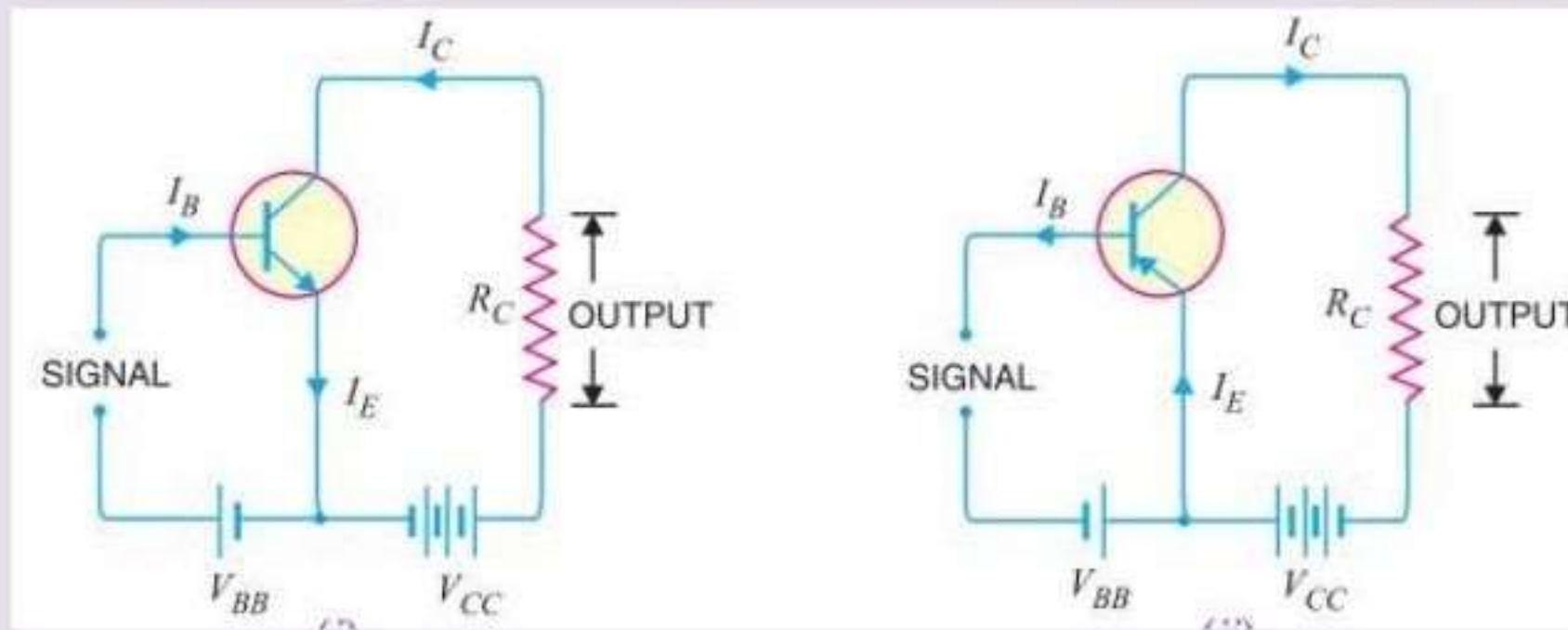
$$r_i = \frac{\Delta V_{BE}}{\Delta I_E}$$

- Output Resistance: The ratio of change in collector-base voltage to the change in collector current is called Output Resistance.

$$r_o = \frac{\Delta V_{BC}}{\Delta I_C}$$

Common Emitter Connection

- The common-emitter terminology is derived from the fact that the emitter is common to both the input and output sides of the configuration.



- First Figure shows common emitter npn configuration and second figure shows common emitter pnp configuration.

Common Emitter Connection

- Base Current amplification factor (β):
- In common emitter connection input current is base current and output current is collector current.
- The ratio of change in collector current to the change in base current is known as base current amplification factor, β .
$$\beta = \frac{\Delta I_C}{\Delta I_B}$$
- Normally only 5% of emitter current flows to base, so amplification factor is greater than 20. Usually this range varies from 20 to 500.

Relation Between β and α

$$\beta = \frac{\Delta I_C}{\Delta I_B}$$

$$\alpha = \frac{\Delta I_C}{\Delta I_E}$$

$$I_E = I_B + I_C$$

$$\Delta I_E = \Delta I_B + \Delta I_C$$

$$\Delta I_B = \Delta I_E - \Delta I_C$$

$$\beta = \frac{\Delta I_C / \Delta I_E}{\frac{\Delta I_E}{\Delta I_E} - \frac{\Delta I_C}{\Delta I_E}} = \frac{\alpha}{1 - \alpha}$$

$$\beta = \frac{\alpha}{1 - \alpha}$$

$$\beta = \frac{\Delta I_C}{\Delta I_E - \Delta I_C}$$

Expression for Collector Current

$$I_C = \alpha I_E + I_{CBO}$$

$$I_E = I_B + I_C = I_B + (\alpha I_E + I_{CBO})$$

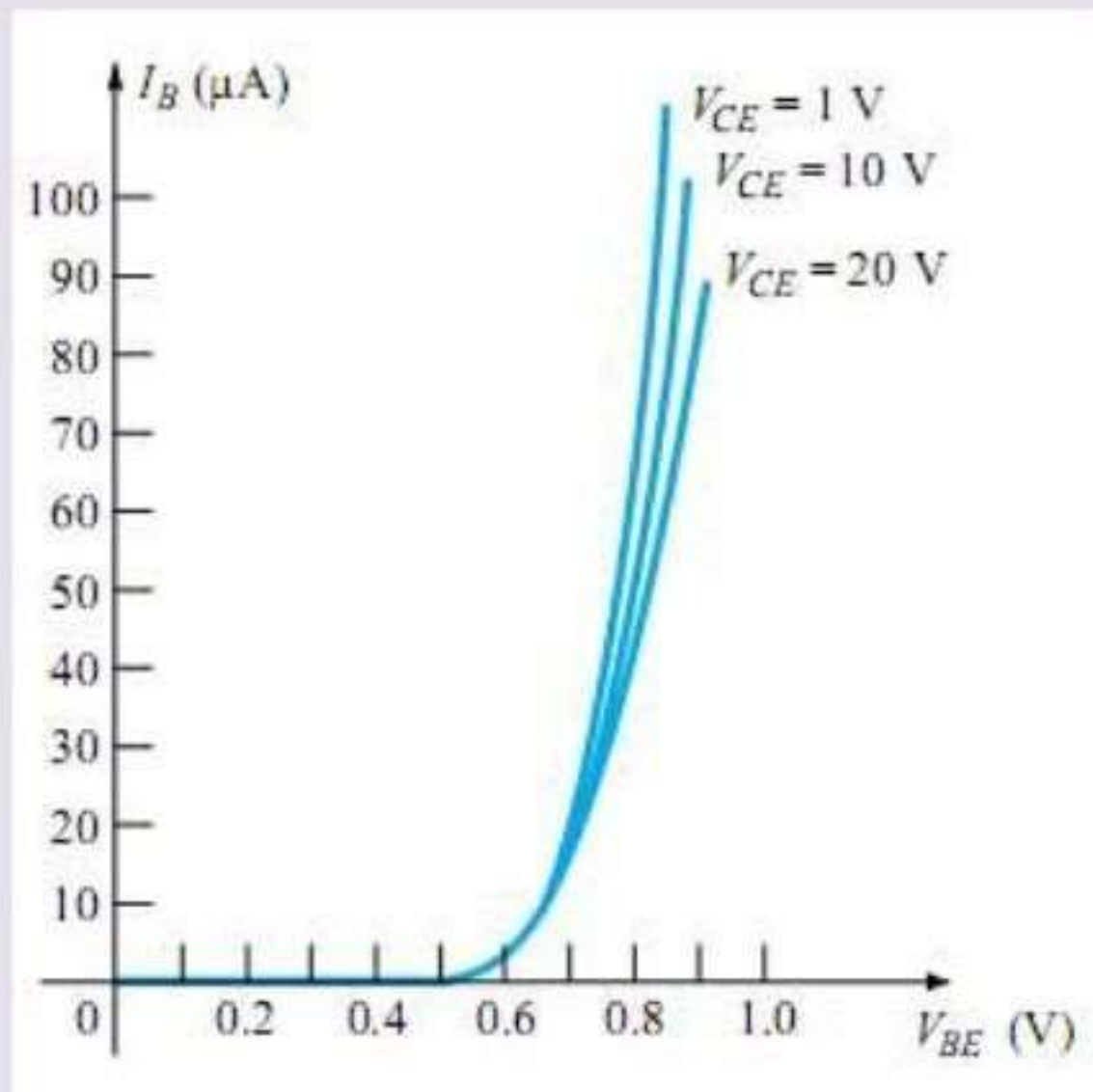
$$I_E (1 - \alpha) = I_B + I_{CBO}$$

$$I_E = \frac{I_B}{1 - \alpha} + \frac{I_{CBO}}{1 - \alpha}$$

$$I_C ; I_E = *(\beta + 1) I_B + (\beta + 1) I_{CBO}$$

Characteristics of common emitter configuration

- Input Characteristics: $\rightarrow V_{BE}$ vs I_B characteristics is called input characteristics.
 - $\rightarrow I_B$ increases rapidly with V_{BE} . It means input resistance is very small.
 - $\rightarrow I_E$ almost independent of V_{CE} .
 - $\rightarrow I_B$ is of the range of micro amps.



Characteristics of common emitter configuration

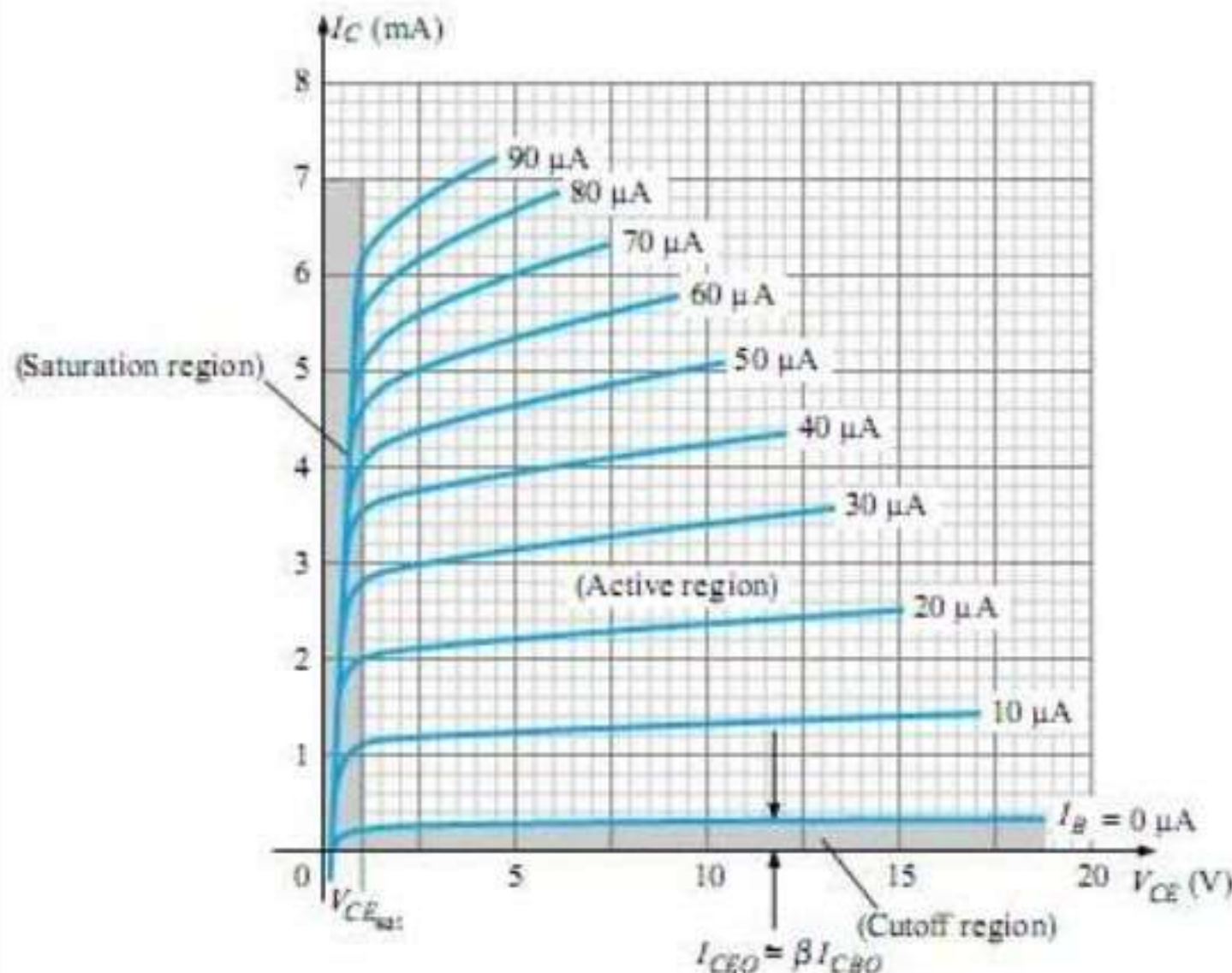
- Output Characteristics:

→ V_{CE} vs I_C

characteristics is called output characteristics.

→ I_C varies linearly with V_{CE} , only when V_{CE} is very small.

→ As, V_{CE} increases, I_C becomes constant.



Input and Output Resistance of common emitter conf.

- Input Resistance: The ratio of change in emitter-base voltage to the change in base current is called Input Resistance.

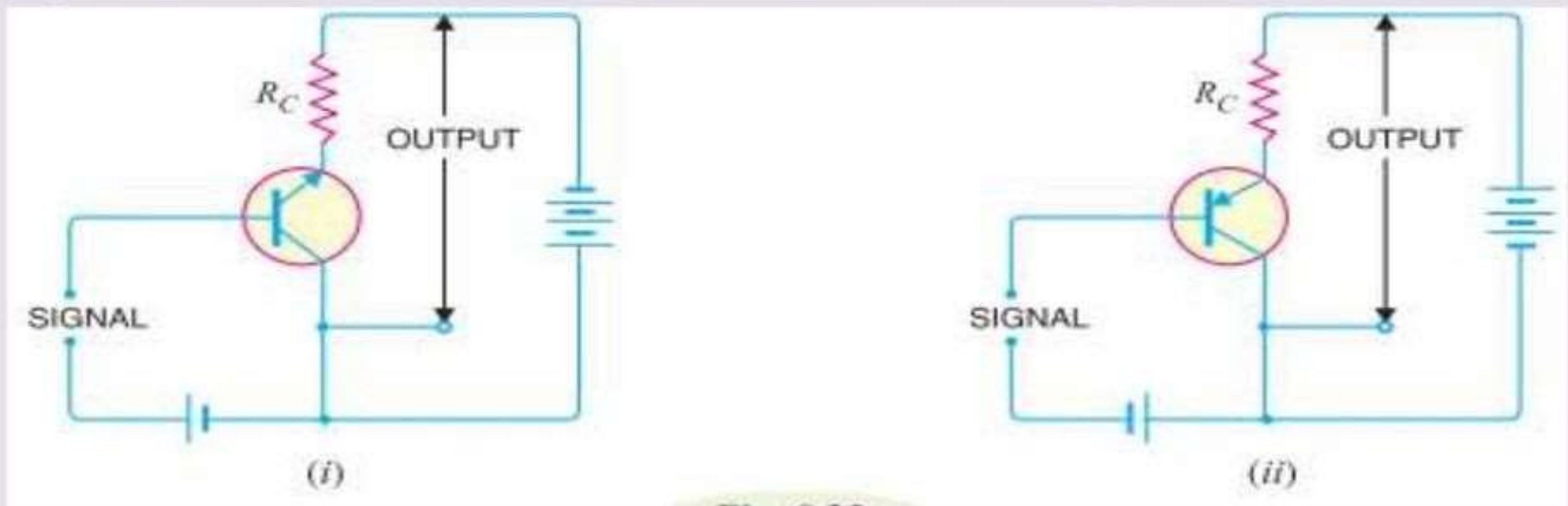
$$r_i = \frac{\Delta V_{BE}}{\Delta I_B}$$

- Output Resistance: The ratio of change in collector-emitter voltage to the change in collector current is called Output Resistance.

$$r_o = \frac{\Delta V_{CE}}{\Delta I_C}$$

Common Collector Configuration

- The common-collector terminology is derived from the fact that the collector is common to both the input and output sides of the configuration.



- First Figure shows common collector npn configuration and second figure shows common collector pnp configuration.

Common Collector Configuration

- Current amplification factor (γ):
- In common emitter connection input current is base current and output current is emitter current.
- The ratio of change in emitter current to the change in base current is known as current amplification factor in common collector configuration.

$$\gamma = \frac{\Delta I_E}{\Delta I_B}$$

- This circuit provides same gain as CE configuration as,

$$\Delta I_E \approx \Delta I_C$$

Relation Between β and α

$$\beta = \frac{\Delta I_E}{\Delta I_B}$$

$$\alpha = \frac{\Delta I_C}{\Delta I_E}$$

$$\beta = \frac{\Delta I_E}{\Delta I_E - \Delta I_C}$$

$$I_E = I_B + I_C$$

$$\Delta I_E = \Delta I_B + \Delta I_C$$

$$\Delta I_B = \Delta I_E - \Delta I_C$$

$$\beta = \frac{\frac{\Delta I_E}{\Delta I_E}}{\frac{\Delta I_E}{\Delta I_E} - \frac{\Delta I_C}{\Delta I_E}} = \frac{1}{1 - \alpha}$$

Expression for Collector Current

$$I_C = \alpha I_E + I_{CBO}$$

$$I_E = I_B + I_C = I_B + (\alpha I_E + I_{CBO})$$

$$I_E (1 - \alpha) = I_B + I_{CBO}$$

$$I_E = \frac{I_B}{1 - \alpha} + \frac{I_{CBO}}{1 - \alpha}$$

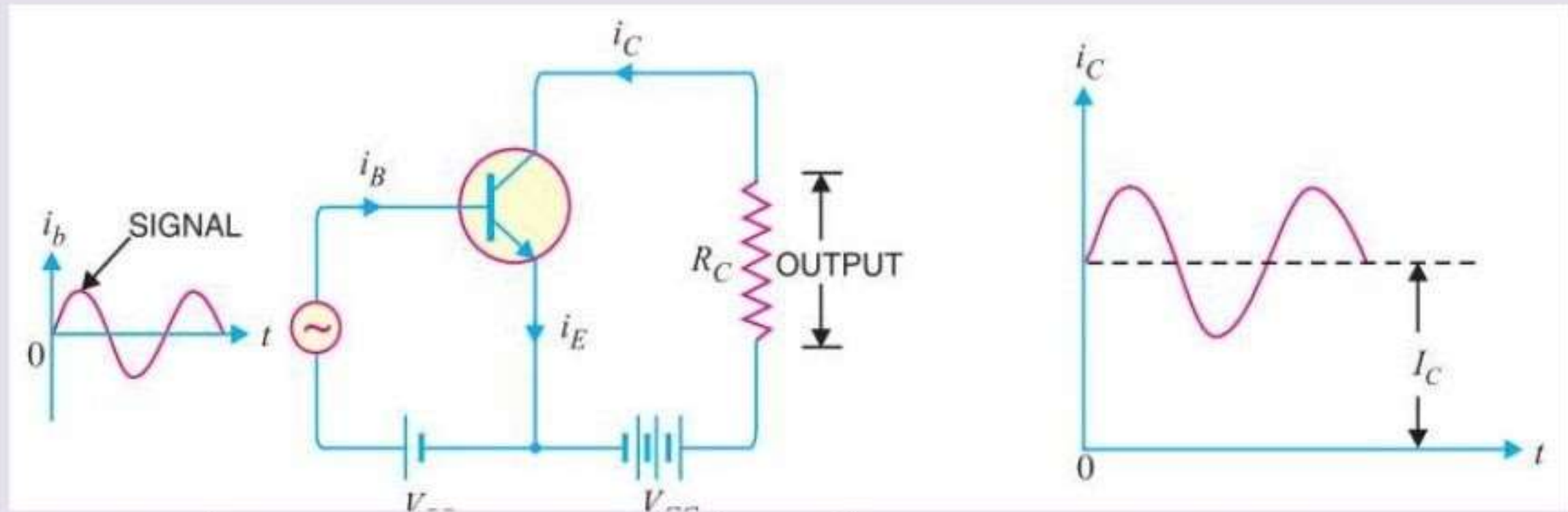
$$I_C ; I_E = *(\beta + 1) I_B + (\beta + 1) I_{CBO}$$

$$\beta = \frac{\alpha}{1 - \alpha} \quad \therefore \quad \beta + 1 = \frac{\alpha}{1 - \alpha} + 1 = \frac{1}{1 - \alpha}$$

Comparison of Transistor Connection

S. No.	Characteristic	Common base	Common emitter	Common collector
1.	Input resistance	Low (about 100 Ω)	Low (about 750 Ω)	Very high (about 750 k Ω)
2.	Output resistance	Very high (about 450 k Ω)	High (about 45 k Ω)	Low (about 50 Ω)
3.	Voltage gain	about 150	about 500	less than 1
4.	Applications	For high frequency applications	For audio frequency applications	For impedance matching
5.	Current gain	No (less than 1)	High (β)	Appreciable

Transistor as an amplifier in CE conf.

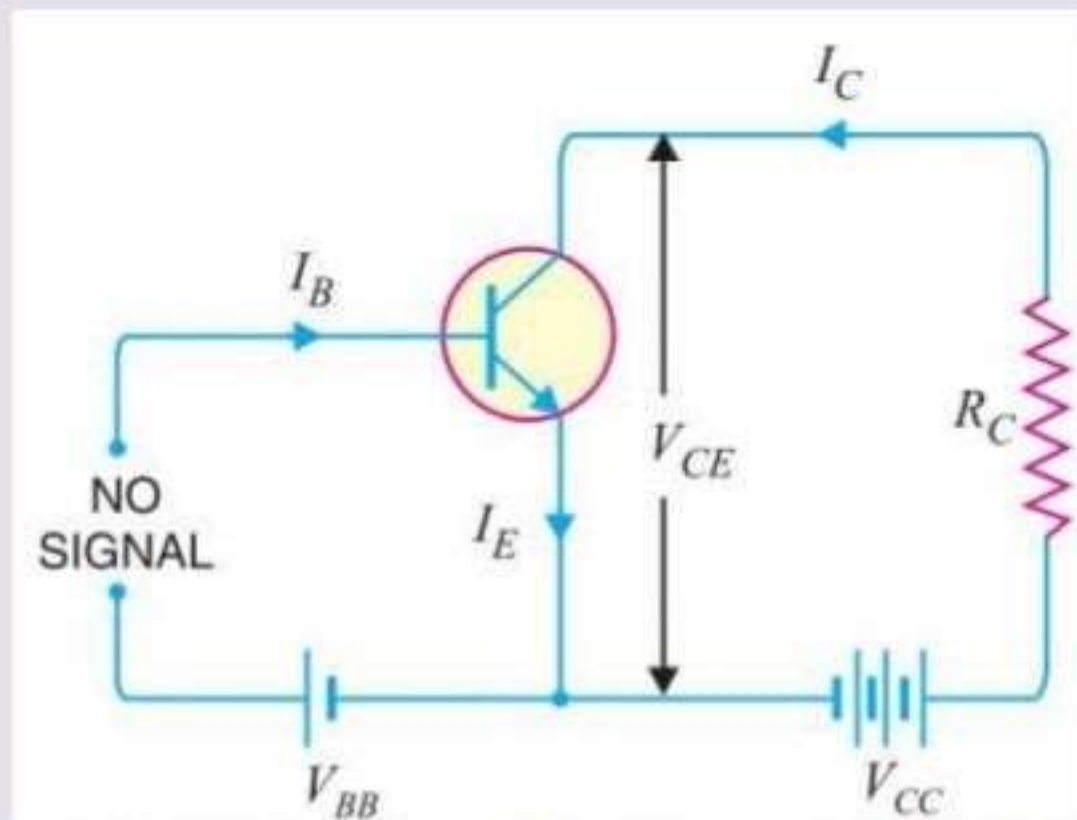


- Figure shows CE amplifier for npn transistor.
- Battery V_{BB} is connected with base in-order to make base forward biased, regardless of input ac polarity.
- Output is taken across Load R

Transistor Load line analysis

- In transistor circuit analysis it is necessary to determine collector current for various V_{CE} voltage.
- One method is we can determine the collector current at any desired V_{CE} voltage, from the output characteristics.
- More conveniently we can use load line analysis to determine operating point.

Transistor Load line analysis



→ Consider common emitter npn transistor ckt shown in figure.

→ There is no input signal.

→ Apply KVL in the output ckt-

$$V_{CE} = V_{CC} - I_C R_C$$

(i) When the collector current $I_C = 0$, then collector-emitter voltage is maximum and is equal to V_{CC} i.e.

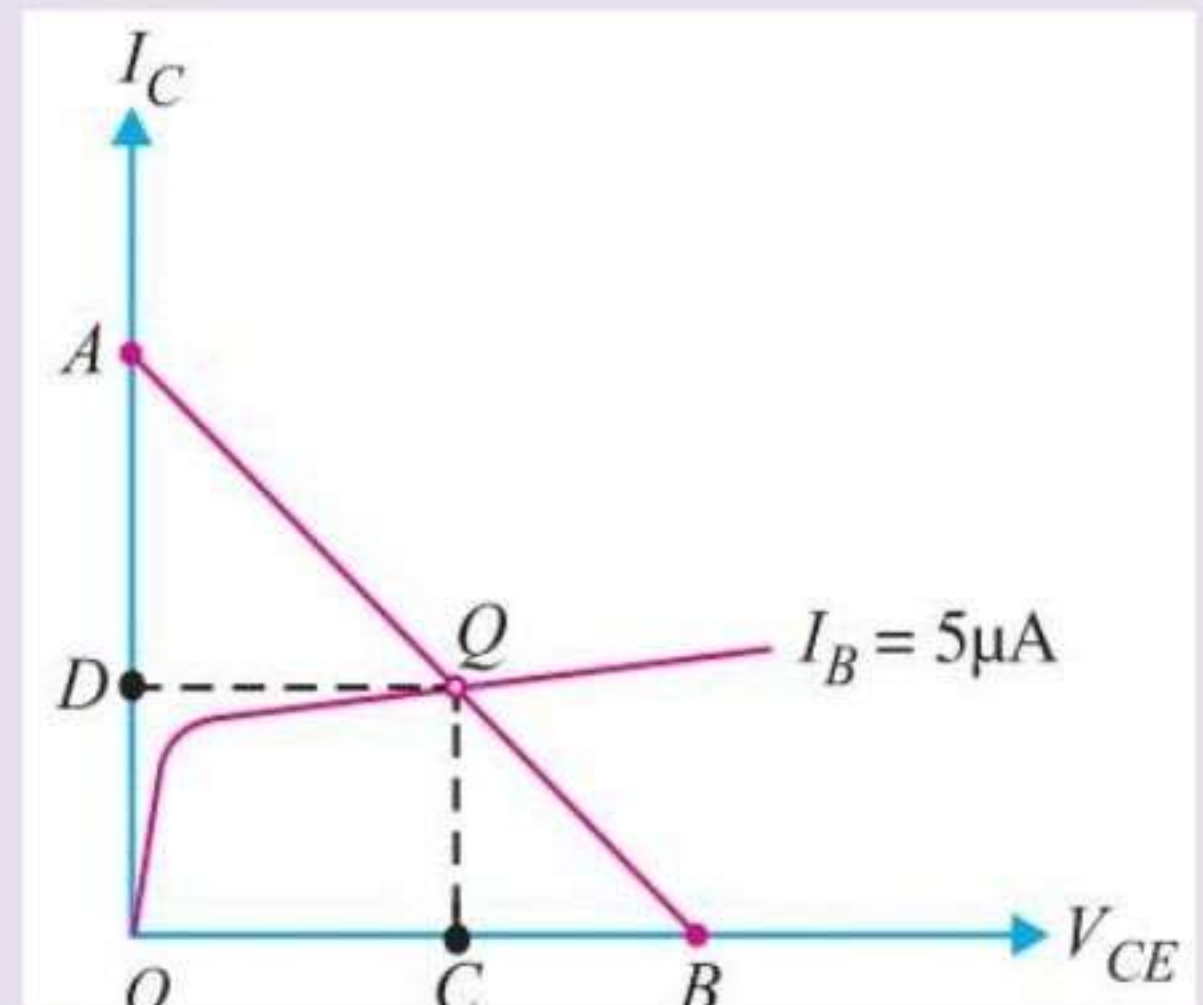
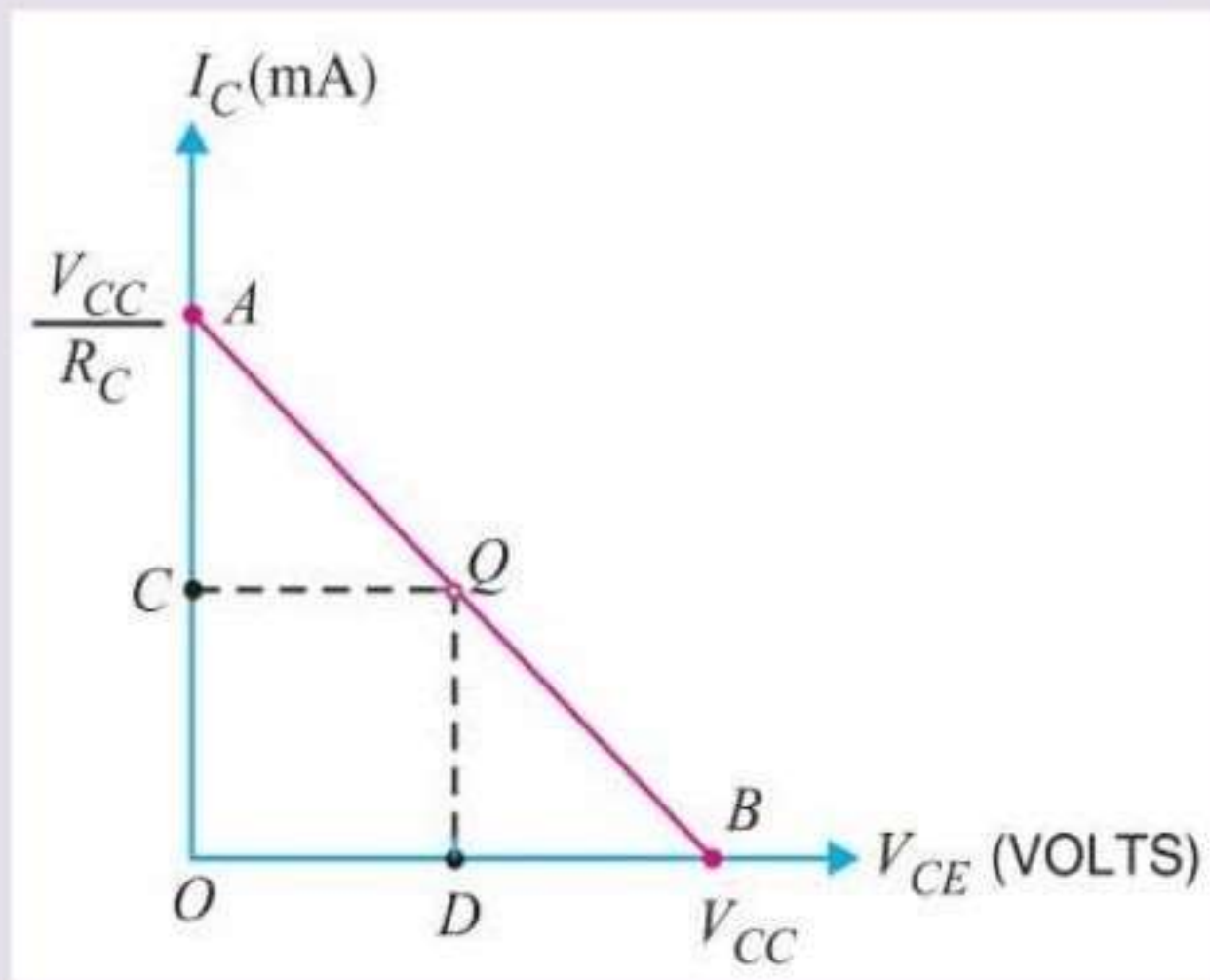
$$\begin{aligned} \text{Max. } V_{CE} &= V_{CC} - I_C R_C \\ &= V_{CC} \quad (\because I_C = 0) \end{aligned}$$

When collector-emitter voltage $V_{CE} = 0$,

$$\begin{aligned} V_{CE} &= V_{CC} - I_C R_C \\ 0 &= V_{CC} - I_C R_C \end{aligned}$$

$$\text{Max. } I_C = V_{CC} / R_C$$

Transistor Load line analysis

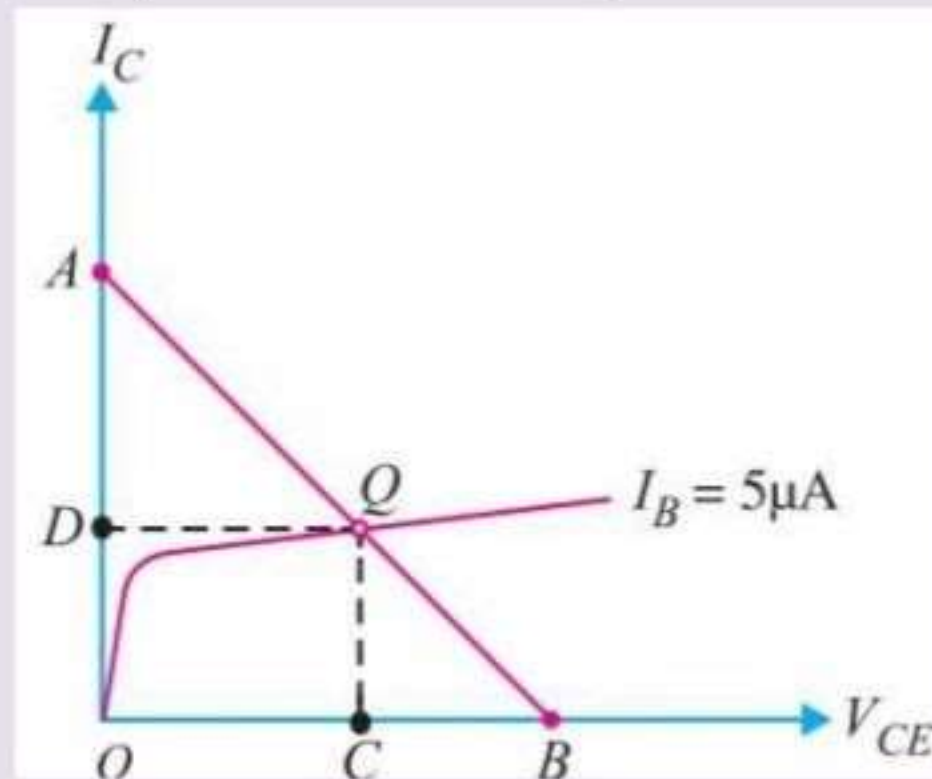


Operating Point

*The zero signal values of I_C and V_{CE} are known as the **operating point**.*

→ It is called operating point because variation of I_C takes place about this point.

→ It is also called quiescent point or Q-point.



ANY QUESTION?

THANKS TO ALL

for stay with me